7B Week 8 Problems

Dobrica Jovanovic

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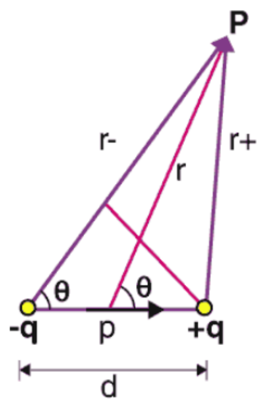
1. A point charge $m = 0.05$kg, with charge $Q = 3$C, is sitting in an external potential $V_i = 17$V. How much work will it take to move the point charge to a point with higher potential $V_f = 22$V.

2. **Deriving dipole electric field:** We can think of an electric dipole $\vec{p}$ as charge $-q$ and charge $+q$ at distance d between each other, such that $\vec{p} = q\vec{d}$ points from negative to positive charge. For electric dipoles we will assume that distance d is much smaller than the other relevant distances in the problem (like distances to other charges for example). We will work in spherical coordinates r, θ, ϕ (although we will only need r and θ for this problem), where we imagine origin as the midpoint of the dipole, and z axis pointing from negative to positive charge (Figure).below



Consider a point P at radius r and angle θ as in the figure. Assume $r >> d$. Denote r_+ and r_- as the distances between P and $+q$ and $-q$ respectively, as in the figure.

(a) Calculate, to the first order in d/r, distances r_+ and r_- in terms of r and θ.
Hint: (Law of cosines) For triangle with side lengths a, b, c if γ is angle opposite to side c, then:

$$c^2 = a^2 + b^2 - 2ab\cos\gamma$$

Hint: (Taylor expansion) For x, a real numbers, such that $x << 1$ we have:

$$(1 + x)^a \approx 1 + ax + O(x^2)$$

(b) Calculate, to the first leading (i.e. nonzero) order in d/r, the electric potential at point P. Notice that your result only depends on q and d as product $p = qd$, which is why we introduce the notion of electric dipole as such! (Hint: use the Taylor expansion)

(c) Using $\vec{E} = -\nabla V$ and the result from the previous part, calculate the electric field at P in spherical coordinates r, θ, ϕ.
Hint: (Polar gradient) Gradient of a function that doesn't depend on ϕ in spherical coordinates is:

$$\nabla f = (\partial_r f)\hat{r} + \left(\frac{1}{r}\partial_\theta f\right)\hat{\theta},$$

where $\hat{r}$ (and respectively $\hat{\theta}$) are unit vectors in the direction of increasing r (respectively θ).

(d) Consider example $\theta = 0$, where P is on the line of the electric dipole. Using Coulomb's law, calculate the electric field from the dipole in the leading order in d/r. Check that your answer agrees with general formula that you found in part (c). (Hint: use the Taylor expansion.).

(e) Consider example $\theta = \frac{\pi}{2}$, i.e. that $\vec{r}$ is orthogonal to $\vec{p}$. Calculate electric field using Coulomb's law and superposition principle and check that your answer agrees with what you have found in part (c). (Hint: This problem appeared in week 5 discussion, problem 3).

3. A particle of mass m, charge $q > 0$ and initial kinetic energy K is projected (from "infinity") towards a heavy nucleus of charge Q, assumed to have a fixed position in our reference frame.

   (a) If the "aim is perfect," how close to the center of the nucleus is the particle when it comes to rest?

   (b) With a particular imperfect aim the particle's closest approach to the nucleus is twice the distance determined in part a. Determine the speed of the particle at this closest distance of approach.

4. Three charges, +5Q,-5Q, and +3Q are located on the y-axis at y = +4a, y = 0, and y = -4a, respectively. The point P is on the x-axis at x = 3a.

   (a) Draw a picture of the situation.

   (b) How much energy did it take to assemble these charges?

   (c) What are the x, y, and z components of the electric field E at P?

   (d) What is the electric potential V at point P, taking V = 0 at infinity?

   (e) A fourth charge of +Q is brought to P from infinity. What are the x, y, and z components of the force F that is exerted on it by the other three charges?

   (f) How much work was done (by the external agent) in moving the fourth charge +Q from infinity to P? This can be done without integrating anything!

5. Suppose we have a charged surface that looks like an empty ice-cream cone. The height of the cone is h and the radius of the base is also h. The surface has a uniform surface charge density σ. Find the potential difference between the tip of the cone and the center of the base. Note that only the sloped surface of the cone is charged, not the base.

   Hint:

$$\int_0^1 \frac{x\,dx}{\sqrt{x^2 + (1-x)^2}} = 1/4(2\sqrt{2x^2 - 2x + 1} - \sqrt{2}\log\left(\sqrt{4x^2 - 4x + 2} - 2x + 1\right))\Big|_0^1 \tag{1}$$

$$= \frac{1}{2\sqrt{2}}\log\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right) \tag{2}$$